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Implications for Gondwana of new Ordovician paleomagnetic data from igneous rocks in southern Victoria Land, East Antarctica

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Abstract. New paleomagnetic data presented here from the southern Victoria Land (SVL) region of East Antarctica further refine the Gondwana early Paleozoic apparent polar wander path. The results are based on paleomagnetic analyses of Early to Middle Ordovician granitoids and dike swarms from which a new SVL pole (23°E, 3.5°S, A95=5.9°) was calculated. The new SVL paleomagnetic pole agrees with less well-determined Ordovician poles from other parts of East Antarctica indicating little or no translation/rotation across the East Antarctic craton since the Middle Ordovician. A new Gondwana ~475 Ma mean pole (11°E, 36°N, A95=7°, N=4 studies, in African coordinates) has been calculated from African, Australian, and Antarctic Early and early Middle Ordovician paleomagnetic poles. Age reassignment of the Antarctic Sør Rondane paleopole (*Zijderveld*, 1968) places it into the Cambrian period and when combined with other Gondwana Cambrian poles results in a new Gondwana ~515 Ma pole (7°E, 22°N, A95=9.5°, N=7 studies, in African coordinates). The new Gondwana ~515 Ma and ~475 Ma poles, when compared with poles of similar age from Laurentia, allow paleogeographic reconstructions to be made that are in keeping with models predicting that Iapetus Ocean basin opening and closure may have been related first to rifting and then collision of Laurentia with Gondwana. The paleomagnetic data also suggest that most of West Gondwana moved toward lower latitudes between the Middle/Late Cambrian and the late Early Ordovician which may be reflected in the fossil record.

Introduction

Antarctica represents one of the key continents within the Gondwana supercontinent yet few paleomagnetic data exist from Antarctica to determine its paleogeographic location during the early Paleozoic. The southern Victoria Land (SVL) area was selected as a good place for paleomagnetic study because it has not experienced any major orogenic event since the Cambro-Ordovician and preliminary paleomagnetic data from both granitoids and dikes yielded apparent early Paleozoic directions consistent with those from Early Ordovician rocks in Africa and Australia [*Luck*, 1972; *Manzoni and Nanni*, 1977; *Funaki*, 1984; *Bachtadse et al.*, 1987].

The objectives of this study were to better define the early Paleozoic Gondwana apparent polar wander path, reassess the Antarctic early Paleozoic paleomagnetic database and create paleogeographic reconstructions of Gondwana testing predictions of divergence and collision of Laurentia and Gondwana [*Dalla Salda et al.*, 1992; *Dalziel et al.*, 1994].

Regional Setting

The rocks in southern Victoria Land (Figure 1) consist of a Proterozoic metamorphic basement of upper amphibolite facies schists and marbles [*Adams and Whitla*, 1991] intruded by Vendian to Ordovician granitoids and dikes [*Faure and Jones*, 1974; *Allibone et al.*, 1991; *Rowell et al.*, 1992]. The metamorphic basement and the early Paleozoic intrusions are overlain unconformably by the flat-lying Beacon Group sedimentary rocks (Devonian to Triassic) [*McKelvey and Webb*, 1962]. All the previously mentioned rock types are cut by ~250-m- thick (and occasionally up to 1000 m) diabase sills and feeder dikes of the Ferrar Group (~175 Ma) [*Allibone et al.*, 1991; *Elliot*, 1991]. Minor late Cenozoic alkaline volcanism (the McMurdo Volcanic Group) has locally occurred along the Ross Sea margin of southern Victoria Land [*Allibone et al.*, 1991].

The paleomagnetic sampling focused on the late Proterozoic to early Paleozoic igneous basement rocks of SVL. Recent mapping has shown the presence of many discrete plutons belonging to three different petrogenetically distinct suites [*Smillie*, 1992; *Allibone et al.*, 1993a]. The Early Ordovician calc-alkalic granitoids are thought to be related to melting associated with westward dipping subduction beneath Antarctica [*Borg and DePaolo*, 1991] and the youngest granitoid suite may relate to uplift and extension in SVL during the Middle Ordovician [*Allibone et al.*, 1993a].

Sampling and Laboratory Procedures

During the austral summer of 1992-1993, 1050 paleomagnetic cores were collected from lower Paleozoic igneous rocks over a 100 x 350 km^2 area in southern Victoria Land [*Grunow and Encarnacion*, 1993]. Samples were taken from equivalent age intrusions over a large region to average out secular variation and evaluate the possibility of structural disruption. Individual tabular plutons were sampled along a

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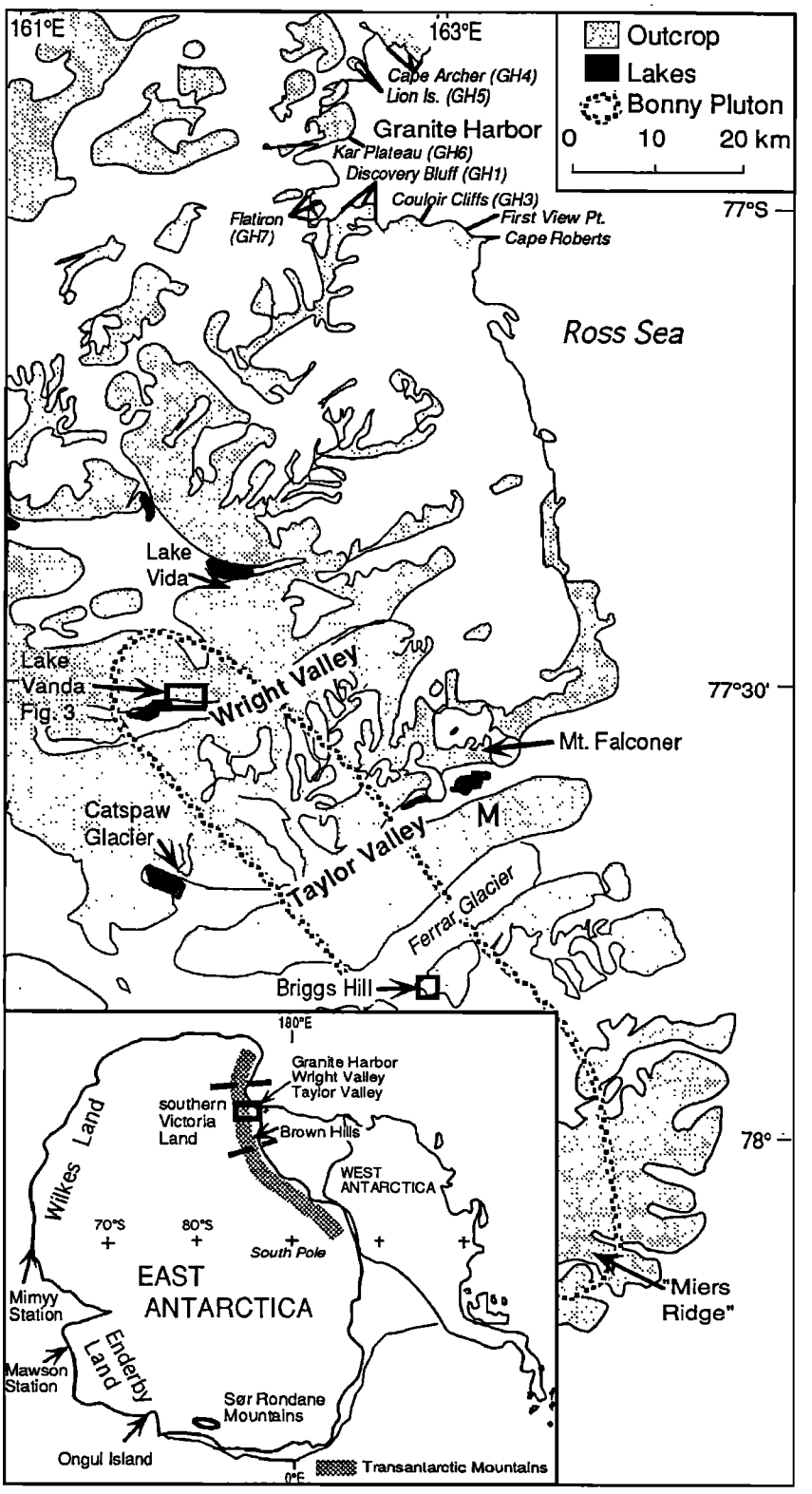


Figure 1. Paleomagnetic locations in southern Victoria Land. M, paleomagnetic study area of *Manzoni and Nanni* [1977]. Location of Bonny pluton from *Cox* [1993]. Inset: Locations of other paleomagnetic studies of early Paleozoic rocks in East Antarctica; box represents location of southern Victoria Land.

60-km strike and across known faults to test the consistency of directions. Generally six to seven cores were sampled at each site using a portable, gasoline-powered, rock-coring drill; four to five samples were collected in dikes less than 1 m wide. The samples were oriented using a magnetic compass. Sun compass measurements were made at each locality to allow correction for local magnetic variation and were always within 2° to 4° of the predicted magnetic variation. Usually four to six sites were collected within each granitoid with sites located at least 100 m apart, and often kilometers apart, to evaluate tilting/rotation of the rocks associated with faulting. The geographic coordinates of each location were determined

from a handheld Global Positioning System (GPS) "Magellan" receiver and are listed to 0.01°in Table 1.

The specimens (cut from each core) were analyzed at the University of Oxford paleomagnetism laboratory. Measurements of the natural remanent magnetization (NRM) were made using a cryogenic magnetometer and/or a computerized fluxgate Molspin magnetometer (housed within Helmholtz coils) on one or more specimens. Pilot samples were progressively demagnetized using stepwise alternating field (AF) or thermal (TH) techniques to determine the best demagnetization procedure for each site. During AF demagnetization, steps of generally 5 to 10 mT up to a peak of 95 mT were used (minimum of eight steps), while during thermal demagnetization steps of generally 50°C or 100°C up to 500°C and then 10° or 20°C intervals up to a maximum of 600°C were used (minimum of 10 steps). Susceptibility measurements were made after each thermal demagnetization step to evaluate mineralogical changes associated with heating the specimen. Isothermal remanent magnetization (IRM) experiments were carried out on representative specimens from the main sample locations. Magnetic saturation was obtained in fields of less than 300 mT and when combined with the observed unblocking temperature spectra suggest that the remanence is carried by titanomagnetite in the sampled intrusions. There is no evidence from either the thermal demagnetization analyses or IRM experiments that hematite is present in the paleomagnetic samples of the igneous rocks described in this paper.

Principal component analysis [*Kirschvink*, 1980] plotted on vector endpoint diagrams was used to determine the magnetization directions for collinear segments converging toward the origin. Directions were considered stable (whether primary or secondary) if defined by three or more collinear points on the vector component diagrams and produced maximum angular deviation values for linear segments (anchored to the origin) of less than 15° (usually less than 7°). For the southern Victoria Land samples, the characteristic remanent magnetization direction (CRM) (determined by the majority of samples displaying a linear decay of the magnetization toward the origin of the vector endpoint diagram) is south to southwest and shallow upward pointing. Mean CRM directions for each locality were obtained by combining sample means to site means (Table 1) and then site means to a unit mean (Table 2) using standard *Fisher* [1953] statistics. Site mean directions with a95 >15° were not used for calculating unit mean directions or virtual geomagnetic poles (VGPs). The regional SVL pole was calculated from the mean poles (calculated by using site-mean VGPs, unless otherwise stated) of the five igneous rock types (Table 2).

Results

The new paleomagnetic data are from basement intrusions at Granite Harbor, Lake Vanda, and Briggs Hill (Figures 1 and 2). The Decade of North American Geology (DNAG) timescale [*Palmer*, 1983] has been used to correlate the isotopic ages of these intrusions to the geologic timescale for all Phanerozoic periods except the Cambrian. For the Cambrian the subdivision by *Bowring et al.* [1993] has been used and suggests that the start of the Middle Cambrian is ~520 Ma.

Baked contact tests were conducted between the mafic and felsic dikes and the country rock granitoids at Granite Harbor and the upper Wright Valley (Figure 1). In all cases no discernible directional difference was found between the dikes and the granitoid bodies and thus results in a negative baked contact test. However, the isotopic data and field relationships indicate that the emplacement of the granites and dikes was in places nearly synchronous which the negative baked contact test reflects.

Paleomagnetic samples were collected from many other areas of southern Victoria Land besides those described below [*Grunow and Encarnacion*, 1993] but (1) yielded erratic directional and intensity changes during demagnetization procedures (as was the case for the Brown Hills granodiorite and the foliated megacrystic granodiorite found on the ridge south of Miers Valley (Figure 1)), or (2) were completely remagnetized by the Ferrar sills at the Catspaw Glacier and the Lake Vida areas (Figure 1), and/or (3) do not yet have isotopic age data to aid in determining the age of magnetic remanence acquisition.

Granite Harbor

Geologists from Sir Robert Falcon Scott's expeditions to Antarctica first noted the different granitoids and lamprophyre dikes in Granite Harbor [*Smith*, 1924] (Figure 1). The paleomagnetic sampling concentrated on two different granitoids with I-type chemistry [*Graham and Palmer*, 1987] intruded by a series of north-northeast trending lamprophyre, feldspar porphyry, and andesitic dikes. A flow-banded, biotite (± hornblende) gray granite with variably aligned, megacrystic orthoclase, was sampled at Discovery Bluff, the Flatiron, Cuff Cape and Kar Plateau (Figure 1). The megacrystic granite at Discovery Bluff has an Rb-Sr biotite age of 476 ± 15 Ma (1σ error) [*Deutsch and Webb*, 1964] (corrected for *Steiger and Jager* [1977] decay constants). A massive, coarse-grained, equigranular, pink biotite-hornblende granite (Suite DV1a), was sampled at Lion Island [*Allibone et al.*, 1993b], eastern Couloir Cliffs, and the Cape Archer area (Figure 1). The Lion Island granite has yielded an Rb-Sr whole rock isochron age of 478 ± 35 Ma (2σ error) [*Allibone et al.*, 1993b]. No contact relationships were observed between the equigranular pink granite and the megacrystic gray granite. Mafic dikes sampled from granite at the Flatiron (Figure 1) display irregular margins, contain enclaves of the host rock granite, and/or occur as stringers within the granite suggestive of nearly synchronous emplacement of the two bodies. Generally, the dikes do not exhibit well-developed chilled margins suggestive of a limited temperature difference between the dikes and the granitoids at the time of dike emplacement.

The five sites drilled in the megacrystic gray granite all yielded characteristic magnetization directions that were upward pointing to the southwest and shallow inclination (Table 1; Figure 2a). Only three of the five sites drilled in the Lion Island granite yielded characteristic directions that were also south directed and shallow (Table 1). The two sites that did not yield stable magnetizations were at Cape Archer where conspicuous color banding may reflect later hydrothermal alteration and at the Kar Plateau where a Ferrar diabase sill is exposed above the granite and has caused chemical alteration of the granite intrusion.

Six dikes (four mafic, one feldspar porphyry and one andesitic dike) all yielded characteristic magnetizations that were south directed and shallow (Table 1; Figure 2b). Three sites were collected from foliated and banded tonalitic-dioritic bodies at the southern end of Lion Island, First View Point and at Cape Roberts (Figure 2) but do not yet have isotopic age information and are not included in this paper.

Table 1. New Southern Victoria Land Site Mean Directions and Virtual Geomagnetic Poles

Site Coordinates Longitude °E, Latitude °S	Site	Isotopic Age	Lithology	Component	NRM x 10^{-2} A/m	Samples, *n/N*	*D*, deg	*I*, deg	*a*95	*k*	VGP Longitude °E	VGP Latitude °N or °S
				Granite Harbor								
162.55, 77.00	GH1A	476±15	grey granite		.012	6/7	219.7	-13.4	8.8	58	22.0	-3.2
162.64, 77.00	GH1B		mafic dike		23.	6/6	211.8	-28.7	5.9	131	13.2	4.2
162.64, 77.00	GH1C	476±15	grey granite		0.67	7/7	219.2	-11.4	5.1	141	21.7	-4.4
162.60, 77.00	GH1D	476±15	grey granite		3.8	7/7	213.8	-22.4	4.7	164	15.6	0.8
162.85, 77.00	GH3A	478±35	pink granite		1.2	7/7	223.7	-1.5	3.8	251	26.9	-8.6
162.85, 76.85	GH4A	478±35	pink granite		2.5	7	--	--	--	--	--	--
162.82, 76.64	GH4B	478±35	pink granite		3.3	7/7	199.5	-26.5	14.2	19	1.7	1.5
162.82, 76.64	GH4C	478±35	pink granite		2.3	6/7	220.5	-12.7	11.6	34	23.0	-3.6
162.59, 76.85	GH5A	478±35	pink granite		2.0	7	--	--	--	--	--	--
162.58, 76.86	GH5C		mafic dike		0.13	5/6	224.0	-12.9	8.8	76	26.4	-3.0
162.56, 76.93	GH6A	476±15	grey granite		2.4	7	--	--	--	--	--	--
162.37, 77.02	GH7A		mafic dike		6.6	5/7	205.7	-16.9	10.9	50	7.8	-3.1
162.42, 77.00	GH7C		andesitic dike		6.7	6/6	208.0	-17.6	12.9	28	10.1	-2.5
162.42, 77.00	GH7D	476±15	grey granite		3.3	7/7	220.5	-6.7	4	231	23.2	-6.5
162.37, 76.99	GH7E		mafic dike		3.9	5/5	215.6	-20.9	6.6	135	17.3	0.2
162.37, 76.99	GH7F		Feldspar porphyritic dike		0.23	3/5	223.7	-19.6	12.7	96	25.2	0.6
162.37, 76.99	GH7G	476±15	grey granite		0.19	7/7	222.9	-10.0	4.5	170	25.3	-4.5
				Lake Vanda								
161.70, 77.52	VN1B	472±15	Bonny pluton		0.7	7/7	215.4	7.0	10.9	32	18.2	-13.6
161.70, 77.52	VN1C	467±15	Feldspar porphyritic dike		1.3	6/6	214.9	-4.4	7.1	91	17.1	-8.0
161.70, 77.52	VN1D	<472±15	grey mafic dike		0.27	4/5	244.7	-34.0	12.6	54	43.3	13.0
161.70, 77.52	VN1E	<472±15	mafic dike		0.52	7	--	--	--	--	--	--
161.74, 77.52	VN1F	<472±15	microdiorite dike		1.2	6/6	220.7	-12.3	9.1	56	22.2	-3.3
161.74, 77.52	VN1G*	<472±15	Sheared felsic dike		0.98	3/5	234.1	-56.8	12.4	100	29.5	29.5
161.70, 77.52	VN1H	472±15	Bonny pluton		0.32	7/8	219.8	-5.9	14.2	23	21.8	-6.6
161.70, 77.52	VN1I	467±15	Feldspar porphyritic dike		0.33	5/6	230.8	2.3	9.4	67	33.5	-9.0
161.70, 77.52	VN1J	467±15	Feldspar porphyritic dike		0.6	6/6	234.3	-11.2	8.6	61	35.7	-1.7
161.70, 77.52	VN1K	467±15	Feldspar porphyritic dike		0.8	6/6	232.1	-13.3	10.5	42	33.4	-1.0
161.70, 77.52	VN1M	467±15	Feldspar porphyritic dike		1.2	7/7	212.5	-11.6	7.6	64	14.1	-4.7
161.70, 77.52	VN1N	467±15	Feldspar porphyritic dike		1.2	5	---	---	---	---	---	---
161.80, 77.52	VN2A	<472±15	Valhalla granite		0.22	6	---	---	---	---	---	---
161.80, 77.52	VN2B	467±15	Feldspar porphyritic dike		0.22	4/5	230.8	-21.0	11.9	61	31.4	2.9
161.80, 77.52	VN2C*	<472±15	Valhalla granite		0.04	4/7	248.5	-78.8	5.9	247	207.6	-61.5
161.78, 77.52	VN2D*	472±15	Bonny pluton		0.16	4/5	227.5	-73.9	14.2	44		
161.78, 77.52	VN2E*	<472±15	Valhalla granite		0.04	4/7	262.2	-78.1	5.4	287	218.8	-62.7
161.68, 77.53	VN3A	472±15	Bonny pluton		0.26	7/7	238.0	-3.6	12.1	26	40.0	-4.8
161.69, 77.53	VN3B	467±15	Feldspar porphyritic dike		0.53	4/6	223.0	-7.5	10.4	79	24.9	-5.4
161.69, 77.53	VN3C	472±15	Bonny pluton		0.19	7/7	228.8	-1.0	11.3	30	31.2	-7.7
				Briggs Hill								
162.98, 77.82	BRG1A	<479 ± 15	mafic dike		13.4	5/6	212.2	-1.9	7.5	105	15.6	-9.3
162.98, 77.82	BRG1B	<479 ± 15	mafic dike		22.4	6/6	202.2	-14.2	8.1	70	5.1	-4.1
162.98, 77.82	BRG1C-E	479 ± 15	Bonny pluton	A	0.07	8/21	239.5	-17.6	14.5	16	41.4	2.7
162.98, 77.82	BRG1C-E*	479 ± 15	Bonny pluton	B	0.07	17/21	298.4	-73.6	6.9	23	259.9	-59.5

Site coordinates determined from handheld GPS Magellan receiver; site locations are shown on Figures 1 and 3; *n/N*, number of samples used for site mean calculation\total number of samples; *D*, in situ site mean declination; *I*, in situ site mean inclination; a95, is the semiangle of the cone of 95% confidence about the site mean; *k*, estimate of precision parameter; VGP, virtual geomagnetic pole; positive values are in northern hemisphere, negative values in southern hemisphere. Present-day field direction for the sample area is Declination ~150°E, Inclination -82°. Isotopic age data are discussed in text.

* Sites not used for Early-Middle Ordovician paleomagnetic pole determination; see text for explanation.

Table 2. Southern Victoria Land Location Mean Directions and Paleopoles From This Study

Lithology	~Age*, Ma	N/*n*	*D*, deg	*I*, deg	*a*95	*k*	PPLon°E	PPLat °S	*A*95, deg	*R*	*dp,dm*	PLAT, deg
					Granite Harbor							
Grey granite	476 ± 15	5/(34)	219.3	-12.8	6.4	146	21.6	-3.6	4.3	4.9876		-6
Pink granite	478 ± 35	3/(21)	214.1	-14.9	8.3	16	16.7	-3.4			4.3,8.5	-8
Felsdspar porphyryr		1/(3)	223.7	-19.6	12.7	96	25.2	0.6			6.9,13.2	-10
Mafic dikes		5/(27)	213.1	-19.5	8.6	80	15.0	0.8	7.5	4.9615		-10
					Lake Vanda							
Bonny pluton	472 ± 15	4/(28)	225.5	-0.9	13.1	50	27.9	-8.3	11.9	3.9505		-4
Mafic dikes		2/(10)	227.1	-17.6	9.3	28	28.1	0.4			5.0,9.7	-9
Felsdspar porphyries	467 ± 15	7/(38)	225.4	-9.6	8.5	51	27.2	-3.9	7.2	6.9166		-5
All Vanda dikes		9/(48)	226.7	-12.3	9.1	33	28.4	-2.1	7.4	8.8382		-6
Mean		13	226.4	-8.8	6.7	34	28.2	-4.0	5.8	12.7729		-4
					Briggs Hill							
Mafic dikes		2/(11)	206.8	-8.6	6.7	47	9.9	-6.5			3.4,6.9	-4
Bonny pluton	479 ± 15	1/(8)	239.5	-17.6	14.5	16	41.4	2.7			7.8, 15	-9
					Lithology Means							
Granite Harbor grey granite	476 ± 15	5/(34)	219.3	-12.8	6.4	146	21.6	-3.6	4.3	4.9876		-6
Granite Harbor pink granite	478 ± 35	3/(21)	214.1	-14.9	8.3	16	16.7	-3.4			4.3,8.5	-8
Mafic dikes	?	9/(48)	215.6	-18.0	9.1	29	17.7	-1.8	7.6	9.7815		-9
Feldspar porphyries	467 ± 15	8/(41)	225.2	-10.9	8.2	55	28.3	-2.6	6.4	6.9338		-5
Bonny pluton	472± 15	5/(36)	228.2	-4.3	13.2	34	30.6	-6.1	11.3	4.9139		-2
					SVL Mean							
Granitoids and dikes	~475 ± 10	5 intrusions	220.8	-11.7	7.6	102	22.7	-3.5	5.9	4.9765		-6

N/n, number of sites used for calculation of mean directions and paleopole/total number of samples represented; underscore, identifies whether sites or samples were used to calculate listed means; *D*, declination; *I*, inclination; a95 error circle at the 95% confidence level; *k*, estimate of precision parameter; PPLon and PPLat, paleopole longitude and latitude; *A*95, radius of error circle at 95% confidence level for the mean paleopole based on the site mean VGPs; *R*, resultant vector; *dp°, dm°*, oval of 95% confidence about the pole position; PLAT, paleolatitude.

* See text for references on isotopic ages. SVL regional mean direction and pole based on mean directions and VGPs of the five igneous intrusions from Granite Harbor grey granite, Granite Harbor pink granite, mafic dikes, feldspar porphyry dikes and the Bonny pluton.

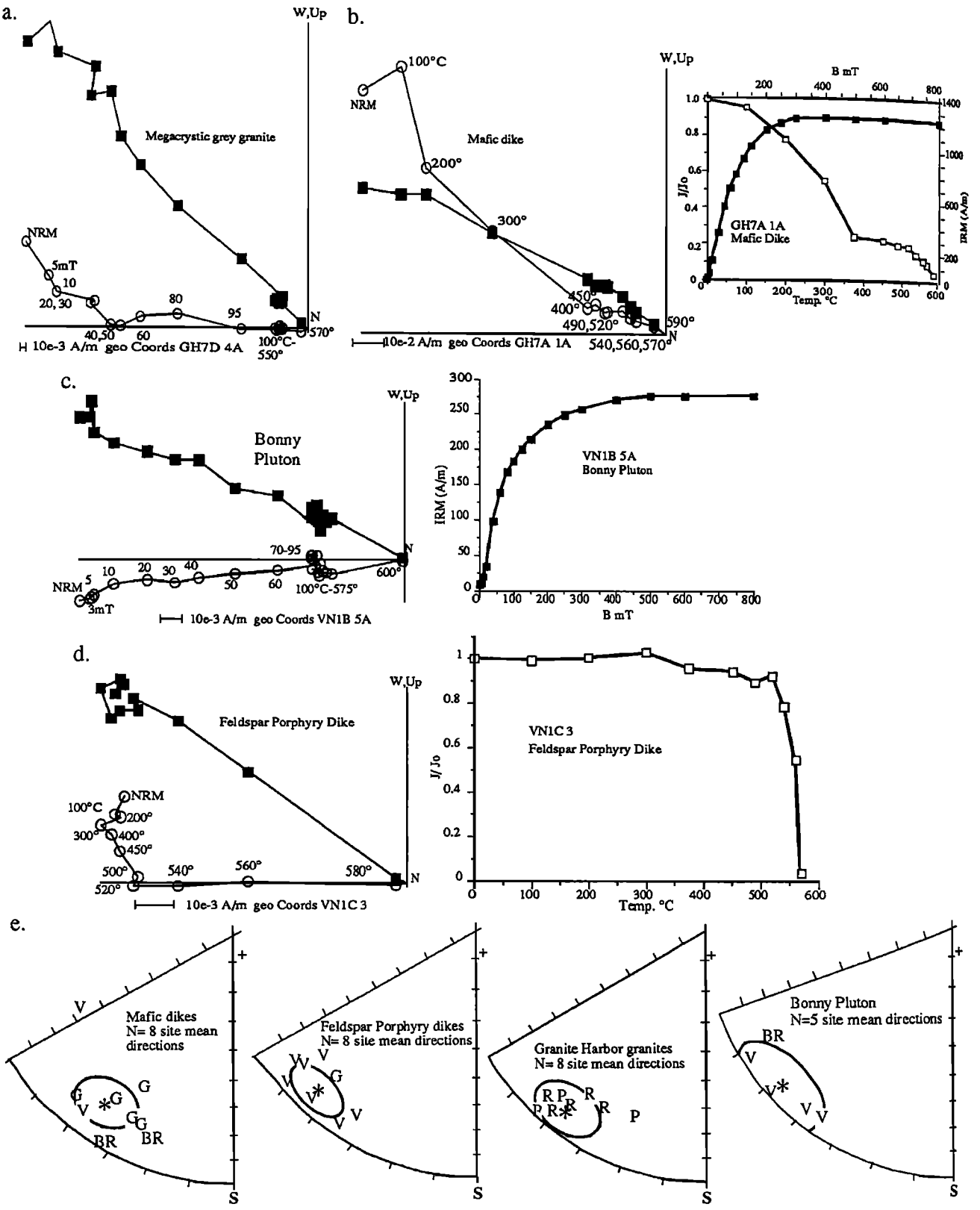


Figure 2. Orthogonal projection of vector endpoint diagrams showing demagnetization behavior of samples from SVL. Open circles (solid squares) are projections on the vertical (horizontal) planes at the indicated levels of AF (millitesla) or thermal (°C) demagnetization. Normalized intensity (J/Jo) plots are shown for lithologies where thermal demagnetization was used. IRM acquisition curves (solid squares) for representative samples are shown; B, direct inductance in millitesla. (a) Granite Harbor granite, (b) Granite Harbor mafic dike, (c) Feldspar porphyry dike at Lake Vanda, d) Bonny pluton at Lake Vanda, (e) equal-angle projections of site mean directions of similar rock types (all symbols are on the upper hemisphere; circle, a95 radius of 95% confidence; asterisk, mean direction; plus, present-day field direction; V, Lake Vanda; G, Granite Harbor; B, Briggs Hill).

Lake Vanda (Upper Wright Valley)

Two intrusions and two lithologically different dike sets were sampled at Lake Vanda in the upper Wright Valley (Figures 1 and 3). The oldest pluton is a coarse grained, megacrystic, granodioritic body, known as the Bonny pluton [*Allibone et al.*, 1991]. The Bonny pluton crops out as a northwest trending, elongate body over a 1000-km^2 area (including Briggs Hill) and is distinctive because of its aligned, megacrystic, alkali-feldspar crystals, the presence of mafic xenoliths and the occurrence of orbicules [*Cox*, 1993]. Flow alignment in the pluton's core is due to magmatic flow before crystallization [*Cox*, 1993]. Rb-Sr isotopic dating of the Bonny pluton in the Wright Valley yields a 472 ± 15 Ma whole rock mineral isochron age (2σ error) [*Allibone et al.*, 1993b].

The Bonny pluton was intruded by an equigranular, medium-grained, biotite granite known as the Valhalla pluton [*Allibone et al.*, 1991]. Intruding both of these plutons are felsic porphyry and aphanitic mafic dikes of the Vanda dike swarm that constitute up to 70% of the outcrop at Lake Vanda. The ~0.5- to ~5-m-wide dikes tend to have sharp, occasionally chilled, contacts. Most of the dikes, regardless of lithology, have a very regular northeast trend, although a few strike northwest. Dikes of both trends were sampled. The geochemistry and mineralogy of the mafic dikes suggest them to be spessartite-type calc-alkaline lamprophyres [*Vennum*, 1990]. The feldspar porphyry dikes (orthoclase and quartz phenocrysts) at Lake Vanda have been dated at 467 ± 15 Ma (1σ error) based on a Rb-Sr hornblende and biotite mineral isochron [*Deutsch and Webb*, 1964] (corrected for *Steiger and Jager* [1977] decay constants) and as 460 ± 7 Ma (1σ error) based on a Rb-Sr whole rock - feldspar mineral isochron [*Jones and Faure*, 1967; *Skinner*, 1983]. The shallow level of emplacement of the Vanda dike swarm and sharp dike contacts suggests that these dikes cooled quickly and that the isotopic ages from them should reflect the time of emplacement.

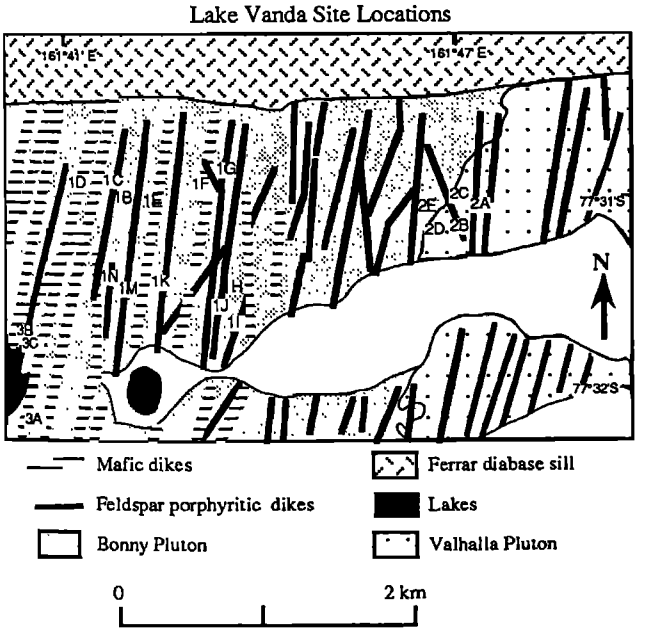


Figure 3. Geologic map [*Allibone et al.*, 1991] and paleomagnetic site locations (Table 1) at Lake Vanda in the Upper Wright Valley.

Pressure estimates indicate that the Bonny pluton was emplaced at pressures of ~5 kbar and the Vanda dike swarm at ~2.5 kbar (based on hornblende geobarometry) [*Allibone et al.*, 1993a] hence requiring some exhumation of the Bonny pluton before the intrusion of the Vanda dike swarm.

Five paleomagnetic sites were collected from the Bonny pluton, four sites from the Valhalla pluton, five sites from the feldspar porphyry dikes, three mafic dikes and three felsic, non-porphyritic dikes (Table 1). The Bonny pluton sites were collected over a 2 x 3 km area and constituted only 30% of the outcrop (the rest being the Vanda dike swarm). The Valhalla pluton sites were collected ~3 km from Lake Vanda (Figure 3). Well-grouped and indistinguishable characteristic remanent directions (south/southwest and shallow) were obtained at high unblocking temperatures (above 450°C) from the Bonny pluton and the Vanda dike sites (Table 1; Figures 2b and 2c). A secondary, stable magnetization was removed below 450°C (southwest and steeply upward pointing) that is similar to the directions found in the Jurassic diabase sills (Declination= 260°, Inclination= -74°).

The Valhalla pluton sites, the one Bonny site adjacent to the intrusive contact with the Valhalla granite (site VN2D) and the feldspar porphyry dike sampled in the Valhalla granite (sites VN2B) all showed a strong Ferrar age (175 Ma) magnetic overprint (Table 1). It is unclear why the Valhalla granite and dikes have been completely remagnetized and/or no longer yield stable magnetic directions given that they do not appear to be nearer to the Ferrar sill (300-m above) than the other sites at Lake Vanda unless there is a Ferrar sill that has stepped upward or a Ferrar feeder dike beneath the Valhalla pluton area. One of the dikes (VN1G) displays a strong structural fabric related to brittle shearing and results from it were not used for any calculations. The magnetic directions from this site are more similar to Middle Jurassic ones (Table 1) and are thought here to reflect remagnetization associated with Jurassic faulting along the dike during emplacement of the Ferrar intrusions. Similar Early Paleozoic dike/Jurassic fault relationships are found throughout SVL [*Wilson*, 1992].

Briggs Hill

Two sites were collected in basaltic andesite dikes and three sites in altered rocks of the Bonny pluton at the western end of Briggs Hill (Figure 1). A Rb-Sr mineral isochron age of 479 ± 15 Ma (2σ error) was obtained from the Bonny pluton 5 km east of Briggs Hill [*Allibone et al.*, 1993b]. The two mafic dikes yielded south and shallow upward-pointing characteristic directions. A west directed, steeply upward-pointing stable magnetic overprint (Declination ~260°, Inclination ~-74°) was removed by ~500°C from the Bonny pluton sites (Table 1) that is indistinguishable from directions expected for the Jurassic Ferrar intrusions (Declination ~255°, Inclination ~-74°) [*Grunow et al.*, 1987] rather than the present day field (Declination =152°, Inclination =-82°) at this location. After removal of the magnetic overprint, about one third of the samples from the weakly magnetized Bonny pluton yielded characteristic directions that were south-southwest directed and shallow-upward pointing (Table 1). As only a few Bonny pluton samples from each site yielded a characteristic magnetization direction, samples from all three sites (n=8) were combined to obtain a Briggs Hill mean direction and pole for the Bonny pluton (Table 2).

Summary of New Data From Southern Victoria Land

Mean directions and poles were calculated from the five different intrusions sampled. A regional mean pole was then determined for southern Victoria Land based on averaging the mean poles from the Granite Harbor grey granite, the Granite Harbor pink granite, the Bonny pluton (from Lake Vanda and Briggs Hill), the mafic dike swarm (from Granite Harbor, Lake Vanda, and Briggs Hill) and the feldspar porphyry dikes (from Lake Vanda and Granite Harbor) (Table 2; Figure 4a).

The intrusion means are considered here a better test for consistency as they represent sites collected from a 40 km x 100 km area within different fault blocks [*Wilson*, 1992]. Major tilting has not occurred in southern Victoria Land since the Devonian, because flat-lying Beacon strata rest unconformably on the granitoids. Tilting may have occurred prior to the Devonian but the consistency of the data over such a large region and within different intrusions of similar age sampled across known structural blocks argues against that. There does not appear to be any significant rotation associated with discrete fault blocks, as the trend of the lower Paleozoic lamprophyre and feldspar porphyry dikes are extremely consistent over a 100 x 300 km region [*Janosy and Wilson*, 1992].

The consistent normal polarity observed in the paleomagnetic data might suggest that either the igneous rocks were emplaced during a relatively short interval of time or that remagnetization has occurred. It is the author's view that remagnetization is unlikely because the paleopoles (1) do not agree with any younger poles from Gondwana; (2) there is no structural deformation or regional orogenic event after ~500 Ma [*Rowell et al.*, 1992; *Cox*, 1993] to possibly result in remagnetization of the igneous rocks; (3) the failed baked contact test of the dikes is not significant because field relationships indicate nearly synchronous emplacement of the dikes and granitoids.

A weighted mean age and error of ~475 ± 10 Ma (2σ error, N = 5 ages using the method of *Taylor* [1982]) has been calculated for the SVL paleomagnetic locations (the Bonny pluton at Lake Vanda and Briggs Hill, the Granite Harbor grey and pink granitoids and the regional feldspar porphyry dike swarm). The mean 475 ± 10 Ma age for the SVL pole places it in the earliest Middle Ordovician with the uncertainty extending the age back to the Early Ordovician (the boundary between the Early and Middle Ordovician is at 478 Ma using the DNAG timescale [*Palmer*, 1983]).

Other East Antarctic Early Paleozoic Paleomagnetic Poles

In general, the Ordovician poles from around East Antarctica agree with each other but almost all lack good isotopic age control. The quality factor, Q (maximum of 7) [*Van der Voo*, 1988] of these studies (Table 3) rate generally from average (3-4) to poor (1-2). Some of the Q factors listed in Table 3 for specific studies differ from those listed by *Van der Voo* [1988] because of new geologic and isotopic data pertaining to the sample areas.

Transantarctic Mountains

A paleomagnetic study on lamprophyre dikes in the lower Taylor Valley of southern Victoria Land (Figure 1) [*Manzoni and Nanni*, 1977] lacks reliable age data and sufficient information to determine which of several dikes sets were sampled. Four sets of igneous dikes are described from the

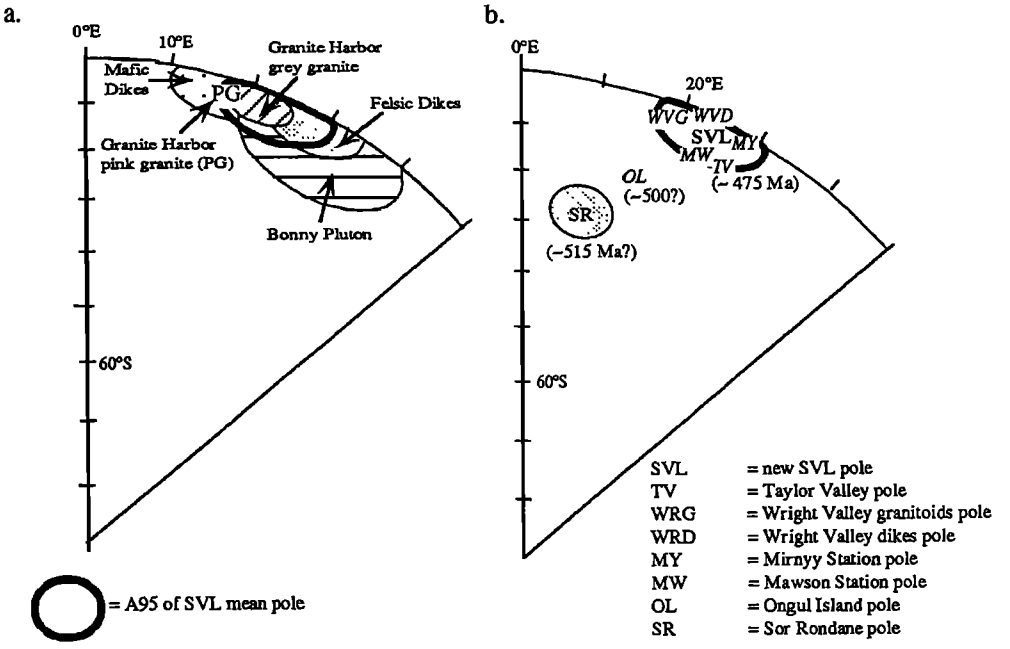


Figure 4. (a) Equal-angle projection of paleomagnetic poles from different rock types with their respective A95 circles (poles are listed in Table 2). BR, Briggs Hill sites; V, Lake Vanda sites; G, Granite Harbor sites; P, pink granite at Granite Harbor; R, grey granite at Granite Harbor. Bold circle represents A95 of mean pole for each igneous intrusion. (b) East Antarctic early Paleozoic paleopoles (listed in Table 3). Almost all of the East Antarctic paleopoles overlap the SVL mean pole (A95 in bold). The Sør Rondane pole [*Zijderveld*, 1968], however, is significantly different from all other East Antarctic poles.

Taylor Valley [*Angino et al.*, 1962] with K-Ar biotite ages of 460 ± 20 and 523 ± 20 Ma [*Pearn et al.*, 1963]. Near the Manzoni and Nanni sample area, a posttectonic, epizonal granite (K-Ar biotite ages of 460 ± 8 Ma [*McDougall and Ghent*, 1970]) cuts the lamprophyre dikes at Mount Falconer (Figure 1) making the lamprophyre dikes older than ~460 Ma. On the basis of geological correlation, the Mount Falconer dikes are probably part of the same mafic dike set sampled by Manzoni and Nanni (Table 3; Figures 1 and 4b).

The new data from Lake Vanda in the Upper Wright Valley corroborate an earlier paleomagnetic study of this area by *Funaki* [1984] (Figure 4b). Recent mapping of plutons in the upper Wright Valley [*Allibone et al.*, 1991] documents the presence of at least four different granitic bodies with different relative emplacement ages. It is uncertain which granitoids were sampled by Funaki.

The East Antarctic Craton

An Early Ordovician pole [*McQueen et al.*, 1972] obtained from charnockitic intrusions dated at 520 ± 24 Ma (2σ error) (Rb-Sr whole rock isochron age adjusted for *Steiger and Jager* [1977] decay constants) at Mirnyy station is based on five sites, three of which have a95's greater than 34° and a fourth site with an a95 of 19° (Table 3; Figures 1 and 4b). Information concerning the temperature at which the characteristic remanent magnetization was acquired is not available. Without additional thermochronometric data, the age of magnetizaton for the Mirnyy Station charnockite is uncertain. Given the large uncertainties associated with the paleomagnetic data and the age of magnetization, this pole cannot be confidently used to represent East Antarctica.

A paleomagnetic study conducted in the Sør Rondane Mountains (Figure 1) [*Zijderveld*, 1968] is based on 16 samples from three different intrusions. One intrusion (Romnaesfjellet/Seal Nunatak) yields U-Pb zircon ages of ~520 and Rb/Sr biotite ages of ~460 Ma [*Picciotto et al.*, 1964]; another intrusion (Lunckerygge) yields an Rb-Sr whole rock age of 525 ± 32 Ma [*Tainosho et al.*, 1992] and an Ar-Ar biotite plateau age of 498 ± 9 Ma [*Takigami et al.*, 1991] while the last location (Vesthaugen) does not yet have published age determinations. Although the age of magnetization for Sør Rondane has usually been considered as 460 to 470 Ma [*Zijderveld*, 1968; *Bachtadse and Briden*, 1990] the new isotopic data suggest that the age should be reconsidered. If uniform cooling of the Lunckerygge granite is assumed, with ~700°C emplacement temperature and an argon biotite closure temperature of ~300°C [*McDougall and Harrison*, 1988] along with a 510°C mean temperature at which the magnetization was acquired (the characteristic magnetization was acquired between 450°C and 575°C [*Zijderveld*, 1968]), then the age of magnetization may be ~515 Ma. This older age provides an explanation for why the Sør Rondane pole is dissimilar to the other East Antarctic Ordovician poles (Table 3; Figure 4b) but similar to the Middle and Late Cambrian poles from Australia (Table 4).

The paleomagnetic data obtained at Ongul Island [*Funaki and Wasilewski*, 1986] and at Mawson Station [*Funaki and Saito*, 1992] in Enderby Land are from granulite facies metamorphic rocks (Table 3; Figures 1 and 4b). The Ongul Island rocks produce a U-Pb ion probe age of 532 ± 6 Ma (1σ error) [*Shiraishi et al.*, 1992] and $^{40}Ar/^{39}Ar$ biotite age of 466 ± 8 Ma (1σ error) [*Takigami et al*, 1992] and hence the likely age of the remanent magnetization is somewhere between ~460 and ~530 Ma. The Mawson Station charnockite yields a U-Pb zircon age of 954 ± 12 Ma [*Young and Black*, 1991] and $^{40}Ar/^{39}Ar$ whole rock analysis yields an age of 444 ± 18 Ma (1σ error) [*Funaki and Saito*, 1992]. The $^{40}Ar/^{39}Ar$ whole rock age is likely to be significantly younger than the age of magnetization because of the low closure temperature of argon in K-feldspar (~150°C) [*McDougall and Harrison*, 1988]. Without more thermochronometric data, these Enderby Land poles are of uncertain value at the present time for plate models or apparent polar wander path compilations.

Comparison With Gondwana Early Paleozoic Poles

It has long been known that the Gondwana Ordovician poles cluster near northern Africa (in an African reference frame) [*Van der Voo*, 1988; *Bachtadse and Briden*, 1990] but the age uncertainties for many of these poles are quite large. The new ~475 Ma early Middle Ordovician SVL paleomagnetic pole is similar to the Early Ordovician poles obtained from the Graafwater Formation in Africa [*Bachtadse et al.*, 1987] and the Australian Jinduckin Formation [*Luck*, 1972] but differs from Middle and Late Ordovician poles from Africa and Australia (Table 4, Figure 5a) [*Embleton*, 1972; *Corner and Henthorn*, 1978; *Bachtadse et al.*, 1987; *Bachtadse and Briden*, 1989]. The isotopic age data from the granitoids used to calculate the SVL pole suggests that the SVL pole cannot be older than ~490 Ma and probably not older than ~480 Ma based on the age of the feldspar porphyry dikes (whose age of primary magnetization and isotopic age should be similar due to rapid cooling of the dikes). If this is the case, then the Ordovician poles from the Graafwater and Jinduckin formations record acquisition of their remanent magnetization during the late Early Ordovician. Combining the Early and early Middle Ordovician poles from Australia, Antarctica, and Africa yields a mean ~475 Ma Gondwana pole of 011E, 36° N, A95=7.2°, N=4 studies (Table 4). If the mafic dikes from Taylor Valley [*Manzoni and Nanni*, 1977] are incorporated with the SVL mafic dike data from this study (assuming they are part of the same dike swarm) then the late Early Ordovician Gondwana pole would be 009°E, 35°N, A95=11°, N=3 continents (Table 4).

There is not good agreement between the Gondwana Middle and Late Ordovician poles (Table 4, Figure 5a). Although the African Middle to Late Ordovician poles are similar to each other, they are significantly different from equivalent age poles from Australia and Antarctica (Figure 5a). The Australian Middle Ordovician Stairway Sandstone pole [*Embleton*, 1972] appears to be anomalous compared with all other Ordovician Gondwana poles (Table 4; Figure 5A). The differences between the Gondwana Ordovician paleomagnetic poles may reflect (1) incorrect age assignments (due to lack of well-determined faunal ages, differences between the faunal and magnetization age, and changes in the geologic timescale); (2) rapid apparent polar wander; (3) unrecognized structural disturbance or remagnetization; (4) differences in demagnetization procedures; (5) motion within Gondwana.

Table 3. Summary of East Antarctic Early Paleozoic Paleomagnetic Poles

Region	Location*	Notes	Lithology	Age, Ma	Number Sites (Samples)	Paleopole Longitude, °E	Paleopole Latitude, °S	A95° (a95°)	Reliability 1	2	3	4	5	6	7	Q	References
southern Victoria Land	Wright Valley, Granite Harbor, Briggs Hill		3 plutons and 17 dikes	~475 ± 10	30 (180)	23°E	4°S	6	X	X	X		X		X	5	this study.
southern Victoria Land	Taylor Valley	1	4 lamprophyre dikes	460 to 520	4 (51)	27°E	9°S	11		X	X		X		X	4	*Manzoni and Nanni* [1977]
southern Victoria Land	Wright Valley	2	lamprophyre dikes	465 ± 15	?(58)	24°E	2°S	[5]	X	X	X		X		X	5	*Funaki* [1984]
southern Victoria Land	Wright Valley	2	granitoids (unspecified)	?	?(16)	18°E	5°S	[8]			X				X	2	*Funaki* [1984]
Enderby Land	Mawson Station	2	charnockite	?	?(88)	22°E	9°S	[2]		X	X				X	3	*Funaki and Saito* [1992]
Enderby Land	Ongul Island	3	gneiss	~500?	3 (65)	17°E	17°S	†17			X				X	2	*Funaki and Wasilewski* [1986]
Queen Maud Land	Sør Rondane Mtns	4	granitoids	~515	3 (16)	10°E	28°S	5			X		X		X	3	*Zijderveld* [1968]
Wilkes Land	Mirnyy Station	5	charnockite	520 ± 24	5 (37)	28°E	2°S	16			X				X	2	*McQueen et al.* [1972]

Reliability criteria based on *Van der Voo* [1988]. A95=radius of error circle at the 95% confidence level for site mean virtual geomagnetic poles; brackets indicate a95 based on sample means; † indicates A95 recalculated by me, Q, quality factor (maximum of 7). The seven criteria are 1, well-determined age; 2, samples >25 and k>10 and α95<16°; 3, demagnetization published in sufficient detail; 4, positive field test (fold, conglomerate, baked contact); 5, tectonic coherence with block or craton and sufficient structural control; 6, presence of antipodal reversals, 7, lack of similarity with poles for younger times. Notes column referes to (1) The dikes have not been dated; dikes elsewhere in Taylor Valley yield K-Ar ages of 460 to 520 Ma. (2) Error difficult to evaluate because number of sites not listed. U-Pb zircon dating yields 1 Ga age, Ar/Ar whole rock yields 445 Ma. Not enough thermochronometric data to determine the age of magnetization. (3) Uncertain what contribution gneissic fabric has on directions. Authors list mean pole based on five sites, I list a mean pole for three sites as one site (EON5) had an a95 of 37° and another site's directional data (WON2) yields a paleopole very different from that listed by the authors. (4) Rb-Sr whole rock and U-Pb zircon ages are ~525 Ma, Ar/Ar biotite plateau ages are ~498, interpolation suggests age of magnetization to be ~515 Ma. (5) Of the five sites, three have site mean α95 values of 32° or greater. Most other charnockites in region are 1 Ga, suggesting isotopic and magnetic? resetting, hence age of magnetization uncertain.

* See Figure 1 for location of paleomagnetic studies.

Table 4. Cambro-Ordovician Gondwana Paleomagnetic Poles in an African Reference Frame

Formation/Unit	Pole	Age	Longitude °E	Latitude °N	A95°	*K*	*Q*	Reference
			Africa					
Cedarberg Formation	1	Upper Ordovician	343	25	9	18	5	*Bachtadse et al.* [1987]
Damara granites	2	458 ± 8 Ma	351	27	5	-		*Corner and Henthorn* [1978]
Salala Ring complex (B)	3	~460 Ma	330	40	9	44	6	*Bachtadse anf Briden* [1989]
Graafwater Formation	4	Lower Ordovician	14	28	9	25	6	*Bachtadse et al.* [1987]
Ntonya	5	522 ± 13 Ma	345	28	2	1045	6	*Briden et al.* [1993]
			Antarctica					
Southern Victoria Land	6	475 ± 10 Ma	8	39	6	171	(5)	this study
Taylor Valley	7	Lower Ordovician	16	37	11	75	4	*Manzoni and Nanni* [1977]
SVL + Taylor Valley	8	~475 ± 10 Ma	8	39	5	225	5	this study and *Manzoni and Nanni* [1977]
Mirnyy Station	9	520 ± 24 Ma	12	44	16	25	(2)	*McQueen et al.* [1972]
Mawson Station	10	?	11	34	-	-	(3)	*Funaki and Saito* [1992]
Ongul Island	11	~500 Ma	12	25	-	-	3	*Funaki andWasilewski* [1986]
Sor Rondane	12	~515 Ma	13	12	5	-	3	*Zijderveld* [1968]
			Australia					
Stairway Sandstone	13	Middle Ordovician	43	46	10	25	3	*Embleton* [1972]
Jinduckin Formation	14	Lower Ordovican	6	38	11	4	-	*Luck* [1972]
Hudson Formation	15	Middle to Upper Cambrian	1	32	13	8	3	*Luck* [1972]
Patapinna Fm.	16	Middle to Upper Cambrian	13	15	17	32	3	*Klootwijk* [1980]
Areyonga	17	Middle Cambrian	358	16	5	574	6	*Klootwijk* [1980]
Lower L. Frome	18	Middle to Upper Cambrian	6	43	14	10	3	*Embleton and Giddings* [1974]
Basal Lake Frome	19	Middle Cambrian	10	22	13	15	5	*Klootwijk* [1980]
Billy Creek	20	Lower toMiddle Cambrian	6	13	14	11	6	*Klootwijk* [1980]
			India					
Salt Beds	21	Middle Cambrian	358	-1	5	176	(4)	*Wensink* [1972]

Mean Gondwana Poles

Time Period	Longitude °E	Latitude °N	A95°	*N*	Continents Represented	Poles Used
~475	11	36	7	4	Australia, Africa, Antarctica	4,6,7,14
~515	7	22	9	7	Antarctica and Australia	12,15,16,17,18,19,20
~515	356	25	25	5	Australia, Antarctica, India	(9,17,18,19,21) *Powell et al.* [1993]

Only poles with *Q* (quality) ratings (see Table 3 for criteria) of 3 or higher with well-determined age constraints have been listed (except the Mirnyy Station pole (9) used by *Powell et al.* [1993] for determining a mean Gondwana pole). Parentheses indicate Q rating different than *Van der Voo* [1990]. Rotation Poles: Australia to Antarctica [*Royer and Sandwell*, 1989]: 38.9°E, 2°S, Δ= -31.5°. Antarctica to Africa [*Norton and Sclater*, 1979]: 327.3°E, 2.4°S, Δ=55.4°. India to Antarctica [*Royer and Sandwell*, 1989]: 10.9°E, 7.8°N, Δ= -65.1°.

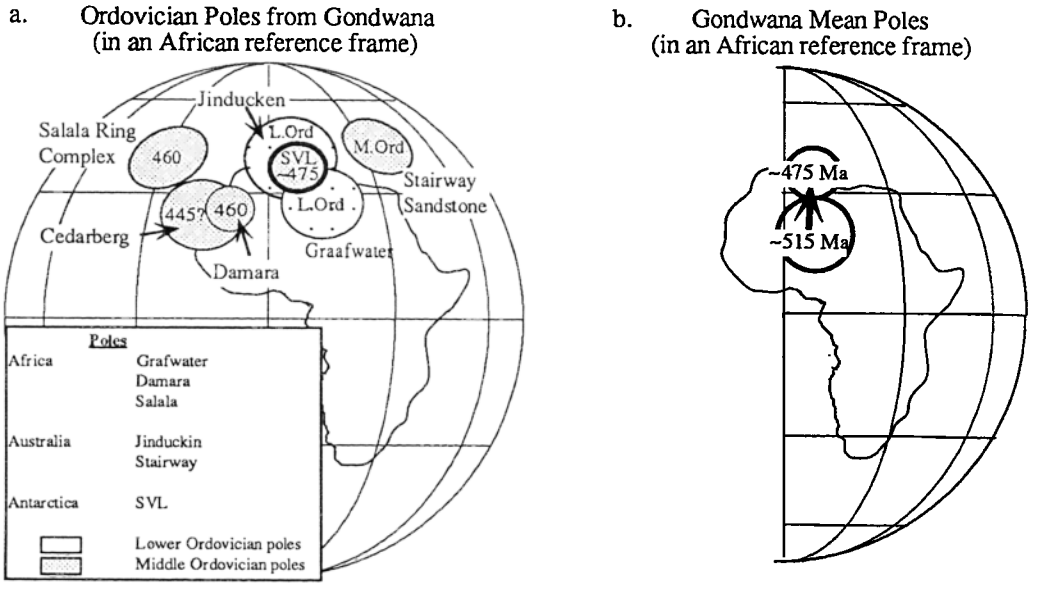


Figure 5. (a) Comparison of Gondwana Ordovician poles (in an African reference frame; see Table 4 for reference and rotation poles used) shows a lack of agreement between supposedly equivalent age Middle Ordovician poles but Early to early-Middle Ordovician poles are similar. The A95 circle for the new Antarctic SVL pole is boldly outlined on this orthographic projection. (b) Comparison of Gondwana Late Cambrian and Early Ordovician poles with their A95 circles (see Table 4) suggests minor apparent polar wander between the ~515 and ~475 Ma.

A new Middle to Late Cambrian East Gondwana pole has been calculated (007E°, 22°N, A95=9°, N=6) which differs from the *Powell et al.* [1993] pole by (1) rejecting the Salt Ranges pole from India because the rocks have undergone rotation associated with India/Asia collision [*Klootwijk et al.*, 1986]; (2) rejecting the Mirnyy Station pole from Antarctica [*McQueen et al.*, 1972] because it is based on poorly defined paleomagnetic and isotopic data; and (3) including the Antarctic Sør Rondane pole and the Australian Patapinna Formation pole [*Klootwijk*, 1980] (Table 4). If the Middle Cambrian starts after ~520 Ma [*Bowring et al.*, 1993] and the Ordovician begins at ~505 Ma (DNAG timescale *Palmer*, 1983]) then the average age of the Middle to Late Cambrian poles from Australia and Antarctica would be about ~510-515 indicate that final assembly of East Gondwana with Africa occurred after the Middle Cambrian [*Grunow et al.*, 1994]. The uncertainty in the Cambrian timescale [*Bowring et al.*, 1993] currently restricts further refinement of the paleogeographic models.

Tectonic Implications

The new Middle to Late Cambrian Gondwana pole (Table 4), considered here to be ~515 Ma, when compared with a mean Cambrian pole from Laurentia [*Van der Voo*, 1990], results in a reconstruction in which a large Iapetus Ocean may have existed between Laurentia and South America (Figure 6a) possibly related to the rifting of Laurentia from Gondwana during the latest Precambrian [*Dalziel et al.*, 1994].

The Gondwana Middle to Late Cambrian pole is based entirely on East Gondwana poles from Australia and Antarctica and differs significantly from the 522 ± 13 Ma African Ntonya complex pole (Table 4) [*Briden et al.*, 1993]. The difference between the East Gondwana and African Cambrian poles could simply reflect apparent polar wander due to the different ages of magnetization represented by the Early Cambrian? African Ntonya pole versus the Middle to Late Cambrian Australian and Antarctic poles (Table 4). Alternatively, if the East Gondwana and Ntonya poles are of similar age, it could

Between ~515 and ~475 Ma, most of Gondwana moved equatorward, especially South America and southern Africa, possibly causing changes in the depositional setting and faunal evolution around parts of Gondwana. The Early Ordovician Gondwana and North America poles [*Jackson and Van der Voo*, 1985] can be interpreted to indicate that eastern North America and western South America were in near proximity, possibly reflecting closure of the Iapetus Ocean (Figure 6b) [*Dalla Salda et al.*, 1992; *Dalziel et al.*, 1994]. Unfortunately, the lack of paleolongitudinal control in paleomagnetic data does not allow exact paleogeographic placement such that North America may not have been located near Gondwana. The possibility that Laurentia was near Baltica based on paleomagnetic and faunal grounds has also been raised [*Torsvik et al.*, 1992]. However, faunal similarities do not necessarily indicate that these continents were at similar latitudes unless the faunas are found in comparable depositional environments, that is, low-latitude, deepwater faunas can be indistinguishable from high-latitude, shallow-water faunas [*Babcock*, 1994]. Hence the lack of paleolongitudinal control from paleomagnetism and the possible paleolatitudinal uncertainty in the faunal data precludes the Baltica/Laurentia or South America/Laurentia models being realized solely by these criteria.

Conclusions

A new late Early Ordovician pole has been determined from the southern Victoria Land region of East Antarctica that is in

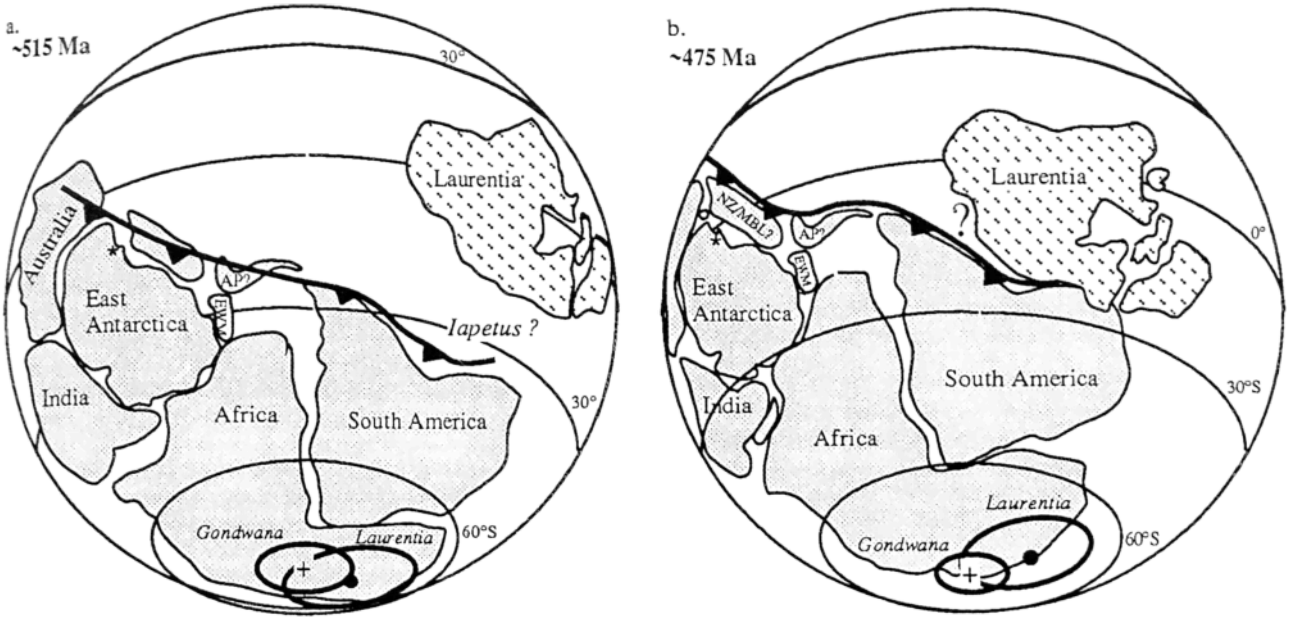


Figure 6. Plate reconstructions of Gondwana (shaded) for ~520 and ~475 Ma using the mean Gondwana poles listed in Table 4 and the Laurentian pole from *Van der Voo* [1990] and *Jackson and Van der Voo* [1985]. The geographic south pole (plus) is centered on the Gondwana reference pole with its A95 circle. The continental fragments of the Antarctic Peninsula (AP), Ellsworth-Whitmore Mountains (EWM) and of New Zealand/Marie Byrd Land (NZ/MBL) contain early Paleozoic rocks [*Spörli*, 1987; *Tangeman et al.*, 1992] and have been speculatively shown in their Mesozoic pre-Gondwana breakup positions relative to East Antarctica [*Grunow*, 1993]. (a) At ~515 Ma. In this reconstruction, the paleomagnetic data allow the possibility that rifting between eastern Laurentia and South America created the Iapetus Ocean [*Dalziel et al.*, 1994]. Emplacement of a major granitic batholith associated with subduction along a convergent margin begins along the Pacificward margin of Antarctica and eastern Australia. (b) At ~475 Ma. Most of Gondwana appears to move equatorward between the Cambrian and Early Ordovician. The paleomagnetic data can be interpreted to be in agreement with the model by *Dalla Salda et al.* [1992] that predicts Ordovician collision of Laurentia and South America (the Taconic and Ocloyic deformational events, respectively) associated with closure of the Iapetus Ocean.

good agreement with Early Ordovician poles from elsewhere in Antarctica and with poles from the other Gondwana continents. The similarity of the early Paleozoic poles around the East Antarctic craton argues for little structural disruption of the rocks in these areas and demonstrates the coherence of East Antarctica since ~475 Ma. The Sør Rondane Antarctic pole [*Zijderveld*, 1968] is considered here to be Middle to Late Cambrian and agrees well with Australian Middle Cambrian poles. New Gondwana ~515 and ~475 Ma paleopoles have been calculated at 007°E, 22°N, A95=9°, N=7 studies and 11°E, 36°N, A95=7°, N=4 studies, respectively. The lack of similarity between the new Gondwana ~515 Ma pole (based on only East Gondwana poles) and the similar age African Ntonya Cambrian pole [*Briden et al.*, 1993] may indicate some motion between East Gondwana and Africa after the Middle Cambrian [*Grunow et al.*, 1994].

The new mean Gondwana Middle to Late Cambrian and Early to early Middle Ordovician poles allow more temporal refinement of paleogeographic and tectonic models. Most of Gondwana appears to have moved toward lower latitudes between ~515 and ~475 Ma (Figures 6a and b) and this may be reflected in the fossil record. Comparison of the ~515 and ~475 Ma Gondwana poles with Laurentian ones can be interpreted to support the idea that the Iapetus Ocean formed as a result of latest Precambrian/Cambrian rifting between Laurentia and Gondwana [*Dalziel et al.*, 1994]. Dalla Salda et al. [1992] suggest that by the Ordovician, closure of the Iapetus Ocean resulted in collision of Laurentia with South America causing the Taconic and Ocloyic orogenies, respectively . The movement of Gondwana toward lower latitudes between ~515 and ~475 Ma may be recording closure of the Iapetus Ocean.

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